

# nRF52 Power Management

## Overview

The nRF52 Power Management module provides battery protection features to prevent over-discharge, minimise likelihood of brownout and flash corruption conditions existing, and enable safe voltage-based recovery.

## Features

### Boot Voltage Protection

- Checks battery voltage immediately after boot and before mesh operations commence
- If voltage is below a configurable threshold (e.g., 3300mV), the device configures voltage wake (LPCOMP + VBUS) and enters protective shutdown (SYSTEMOFF)
- Prevents boot loops when battery is critically low
- Skipped when external power (USB VBUS) is detected

### Voltage Wake (LPCOMP + VBUS)

- Configures the nRF52's Low Power Comparator (LPCOMP) before entering SYSTEMOFF
- Enables USB VBUS detection so external power can wake the device
- Device automatically wakes when battery voltage rises above recovery threshold or when VBUS is detected

### Early Boot Register Capture

- Captures RESETREAS (reset reason) and GPREGRET2 (shutdown reason) before SystemInit() clears them
- Allows firmware to determine why it booted (cold boot, watchdog, LPCOMP wake, etc.)
- Allows firmware to determine why it last shut down (user request, low voltage, boot protection)

### Shutdown Reason Tracking

Shutdown reason codes (stored in GPREGRET2):

| Code | Name         | Description                           |
|------|--------------|---------------------------------------|
| 0x00 | NONE         | Normal boot / no previous shutdown    |
| 0x4C | LOW_VOLTAGE  | Runtime low voltage threshold reached |
| 0x55 | USER         | User requested powerOff()             |
| 0x42 | BOOT_PROTECT | Boot voltage protection triggered     |

## Supported Boards

| Board                                   | Implemented | LPCOMP wake | VBUS wake |
|-----------------------------------------|-------------|-------------|-----------|
| Seeed Studio XIAO nRF52840 (xiao_nrf52) | Yes         | Yes         | Yes       |
| RAK4631 (rak4631)                       | Yes         | Yes         | Yes       |
| Heltec T114 (heltec_t114)               | Yes         | Yes         | Yes       |
| GAT562 Mesh Watch13                     | Yes         | Yes         | Yes       |
| Promicro nRF52840                       | No          | No          | No        |
| RAK WisMesh Tag                         | No          | No          | No        |

| Board                           | Implemented | LPCOMP wake | VBUS wake |
|---------------------------------|-------------|-------------|-----------|
| Heltec Mesh Solar               | No          | No          | No        |
| LilyGo T-Echo / T-Echo Lite     | No          | No          | No        |
| SenseCAP Solar                  | Yes         | Yes         | Yes       |
| WIO Tracker L1 / L1 E-Ink       | No          | No          | No        |
| WIO WM1110                      | No          | No          | No        |
| Mesh Pocket                     | No          | No          | No        |
| Nano G2 Ultra                   | No          | No          | No        |
| ThinkNode M1/M3/M6              | No          | No          | No        |
| T1000-E                         | No          | No          | No        |
| Ikoka Nano/Stick/Handheld (nRF) | No          | No          | No        |
| KeepTeen LT1                    | No          | No          | No        |
| Minewsemi ME25LS01              | No          | No          | No        |

Notes: - "Implemented" reflects Phase 1 (boot lockout + shutdown reason capture). - User power-off on Heltec T114 does not enable LPCOMP wake. - VBUS detection is used to skip boot lockout on external power, and VBUS wake is configured alongside LPCOMP when supported hardware exposes VBUS to the nRF52.

## Technical Details

### Architecture

The power management functionality is integrated into the `NRF52Board` base class in `src/helpers/NRF52Board.cpp`. Board variants provide hardware-specific configuration via a `PowerMgtConfig` struct and override `initiateShutdown(uint8_t reason)` to perform board-specific power-down work and conditionally enable voltage wake (LPCOMP + VBUS).

### Early Boot Capture

A static constructor with priority 101 in `NRF52Board.cpp` captures the `RESETREAS` and `GPREGRET2` registers before: - `SystemInit()` (priority 102) - which clears `RESETREAS` - Static C++ constructors (default priority 65535)

This ensures we capture the true reset reason before any initialisation code runs.

### Board Implementation

To enable power management on a board variant:

1. **Enable in `platformio.ini`:** `ini -D NRF52_POWER_MANAGEMENT`
2. **Define configuration in `variant.h`:** `c #define PWRMGT_VOLTAGE_BOOTLOCK 3300 // Won't boot below this voltage (mV) #define PWRMGT_LPCOMP_AIN 7 // AIN channel for voltage sensing #define PWRMGT_LPCOMP_REFSEL 2 // REFSEL (0-6=1/8..7/8, 7=ARef, 8-15=1/16..15/16)`
3. **Implement in `board.cpp` file:**

```
#ifndef NRF52_POWER_MANAGEMENT const PowerMgtConfig power_config =
{ .lpcomp_ain_channel = PWRMGT_LPCOMP_AIN, .lpcomp_refsel =
PWRMGT_LPCOMP_REFSEL, .voltage_bootlock = PWRMGT_VOLTAGE_BOOTLOCK };
```

```

void MyBoard::initiateShutdown(uint8_t reason)
{
    // Board-specific shutdown preparation (e.g., disable peripherals) bool
    enable_lpcomp = (reason == SHUTDOWN_REASON_LOW_VOLTAGE || reason ==
SHUTDOWN_REASON_BOOT_PROTECT);

    if (enable_lpcomp) {
        configureVoltageWake(power_config.lpcomp_ain_channel,
power_config.lpcomp_refsel);
    }

    enterSystemOff(reason);
} #endif

void MyBoard::begin()
{
    NRF52Board::begin(); // or NRF52BoardDCDC::begin() // ... board setup ...
#ifdef NRF52_POWER_MANAGEMENT checkBootVoltage(&power_config);
#endif
}

```

For user-initiated shutdowns, `powerOff()` remains board-specific. Power management only arms LPCOMP for automated shutdown reasons (boot protection/low voltage).

#### 1. Declare override in board .h file:

```

#ifdef NRF52_POWER_MANAGEMENT

void initiateShutdown(uint8_t reason) override;

#endif

```

## Voltage Wake Configuration

The LPCOMP (Low Power Comparator) is configured to: - Monitor the specified AIN channel (0-7 corresponding to P0.02-P0.05, P0.28-P0.31) - Compare against VDD fraction reference (REFSEL: 0-6=1/8..7/8, 7=ARef, 8-15=1/16..15/16) - Detect UP events (voltage rising above threshold) - Use 50mV hysteresis for noise immunity - Wake the device from SYSTEMOFF when triggered

VBUS wake is enabled via the POWER peripheral USBDETECTED event whenever `configureVoltageWake()` is used. This requires USB VBUS to be routed to the nRF52 (typical on nRF52840 boards with native USB).

#### LPCOMP Reference Selection (PWRMGT\_LPCOMP\_REFSEL):

| REFSEL | Fraction | VBAT @ 1M/1M divider (VDD=3.0-3.3) | VBAT @ 1.5M/1M divider (VDD=3.0-3.3) |
|--------|----------|------------------------------------|--------------------------------------|
| 0      | 1/8      | 0.75-0.82 V                        | 0.94-1.03 V                          |
| 1      | 2/8      | 1.50-1.65 V                        | 1.88-2.06 V                          |
| 2      | 3/8      | 2.25-2.47 V                        | 2.81-3.09 V                          |
| 3      | 4/8      | 3.00-3.30 V                        | 3.75-4.12 V                          |
| 4      | 5/8      | 3.75-4.12 V                        | 4.69-5.16 V                          |
| 5      | 6/8      | 4.50-4.95 V                        | 5.62-6.19 V                          |
| 6      | 7/8      | 5.25-5.77 V                        | 6.56-7.22 V                          |
| 7      | ARef     | -                                  | -                                    |
| 8      | 1/16     | 0.38-0.41 V                        | 0.47-0.52 V                          |

| REFSEL | Fraction | VBAT @ 1M/1M divider (VDD=3.0-3.3) | VBAT @ 1.5M/1M divider (VDD=3.0-3.3) |
|--------|----------|------------------------------------|--------------------------------------|
| 9      | 3/16     | 1.12-1.24 V                        | 1.41-1.55 V                          |
| 10     | 5/16     | 1.88-2.06 V                        | 2.34-2.58 V                          |
| 11     | 7/16     | 2.62-2.89 V                        | 3.28-3.61 V                          |
| 12     | 9/16     | 3.38-3.71 V                        | 4.22-4.64 V                          |
| 13     | 11/16    | 4.12-4.54 V                        | 5.16-5.67 V                          |
| 14     | 13/16    | 4.88-5.36 V                        | 6.09-6.70 V                          |
| 15     | 15/16    | 5.62-6.19 V                        | 7.03-7.73 V                          |

**Important:** For boards with a voltage divider on the battery sense pin, LPCOMP measures the divided voltage.

Use:  $\text{VBAT\_threshold} \approx (\text{VDD} * \text{fraction}) * \text{divider\_scale}$ , where  $\text{divider\_scale} = (\text{Rtop} + \text{Rbottom}) / \text{Rbottom}$  (e.g., 2.0 for 1M/1M, 2.5 for 1.5M/1M, 3.0 for XIAO).

## SoftDevice Compatibility

The power management code checks whether SoftDevice is enabled and uses the appropriate API: - When SD enabled: `sd_power_*` functions - When SD disabled: Direct register access (NRF\_POWER->\*)

This ensures compatibility regardless of BLE stack state.

## CLI Commands

Power management status can be queried via the CLI:

| Command                            | Description                                                           |
|------------------------------------|-----------------------------------------------------------------------|
| <code>get pwrmtg.support</code>    | Returns "supported" or "unsupported"                                  |
| <code>get pwrmtg.source</code>     | Returns current power source - "battery" or "external" (5V/USB power) |
| <code>get pwrmtg.bootreason</code> | Returns reset and shutdown reason strings                             |
| <code>get pwrmtg.bootmv</code>     | Returns boot voltage in millivolts                                    |

On boards without power management enabled, all commands except `get pwrmtg.support` return:

ERROR: Power management not supported

## Debug Output

When `MESH_DEBUG=1` is enabled, the power management module outputs:

```
DEBUG: PWRMGT: Reset = Wake from LPCOMP (0x200000); Shutdown = Low Voltage (0x4C)
DEBUG: PWRMGT: Boot voltage = 3450 mV (threshold = 3300 mV)
DEBUG: PWRMGT: LPCOMP wake configured (AIN7, ref=3/8 VDD)
```

## Phase 2 (Planned)

- Runtime voltage monitoring
- Voltage state machine (Normal -> Warning -> Critical -> Shutdown)
- Configurable thresholds
- Load shedding callbacks for power reduction
- Deep sleep integration
- Scheduled wake-up
- Extended sleep with periodic monitoring

## References

- [nRF52840 Product Specification - POWER](#)
- [nRF52840 Product Specification - LPCOMP](#)
- [SoftDevice S140 API - Power Management](#)